UNIT 1 – INTRODUCTION TO CYBER CRIME

INTRODUCTION TO CYBER CRIME : Cybercrime- Definition and Origins of the word Cybercrime and Information Security, Who are Cybercriminals? Classifications of Cybercrimes, A Global Perspective on Cybercrimes, Cybercrime Era: Survival Mantra for the Netizens. Cyber offenses: How Criminals Plan the Attacks, Social Engineering, Cyber stalking, Cybercafe and Cybercrimes, Botnets: The Fuel for Cybercrime, Attack Vector.

1 INTRODUCTION TO CYBERCRIME

1.1 Meaning of Cybercrime

Cybercrime refers to illegal activities performed using:

- Computer
- Network
- Internet
- Digital devices

It can be:

1. Computer as a target
2. Computer as a tool
3. Computer as both

1.2 Characteristics of Cybercrime

1. Borderless (No geographical limit)
2. Anonymous
3. Fast execution
4. Hard to trace
5. Low cost but high impact

Example:
A hacker sitting in another country can attack Indian bank servers remotely.

1.3 Origin of the Term "Cyber"

- Derived from "Cybernetics"
- Popular after internet growth (1990s)
- As digital communication increased → digital crimes increased

2 INFORMATION SECURITY

Information Security means protecting digital information from:

- Unauthorized access
- Misuse
- Destruction
- Disclosure

2.1 Objectives of Information Security (CIA Triad)

A. Confidentiality

Definition: Data must be accessible only to authorized users.

Technical Methods:

- Encryption (AES, RSA)
- Password protection
- Access control lists

Example:

Bank account details must not be visible to unauthorized person.

B. Integrity

Definition: Data must remain accurate and unchanged.

Technical Methods:

- Hashing (SHA-256)
- Digital signatures
- Version control

Example:

Marks stored in university database should not be altered.

C. Availability

Definition: Data must be available whenever required.

Methods:

- Backup servers
- Redundant systems
- Cloud hosting

Example:

Online exam portal must work during exam time.

3 WHO ARE CYBERCRIMINALS?

Cybercriminals are individuals or groups who exploit digital systems for illegal gain.

3.1 Types of Cybercriminals

1. Script Kiddies

- Low skill
- Use ready-made tools

Example:
Downloading hacking tools from internet and attacking websites.

2. Professional Hackers

- Highly skilled
- Financial motivation

Example:
Bank fraud operations.

3. Insider Threats

- Employees inside organization

Example:
Employee leaking company data.

4. Hacktivists

- Political or social motive

Example:
Defacing government website.

5. Cyber Terrorists

- National security target

Example:
Attacking power grid.

4️⃣ CLASSIFICATION OF CYBERCRIMES

Cybercrimes are classified based on target.

4.1 Crimes Against Individuals

1. Identity Theft
2. Cyber Stalking
3. Email Spoofing
4. Phishing

5. Online Harassment

Example:
Stealing someone's Aadhaar details for fraud.

4.2 Crimes Against Property

1. Intellectual Property Theft
2. Ransomware
3. Data theft
4. Credit card fraud

Example:
Encrypting company data and asking ransom.

4.3 Crimes Against Organizations

1. DDoS attacks
2. Database hacking
3. Website defacement
4. Corporate espionage

Example:
Taking down e-commerce website during sale.

4.4 Crimes Against Government

1. Cyber warfare
2. Spying
3. Critical infrastructure attack

Example:
Attacking railway control system.

5️⃣ GLOBAL PERSPECTIVE ON CYBERCRIME

Cybercrime is a global issue because:

- Internet connects all countries.
- Cryptocurrency allows anonymous payments.
- Law enforcement coordination is difficult.

Global Impact

- Financial loss (billions annually)

- National security threats
- Data privacy violations

6️⃣ CYBERCRIME ERA – SURVIVAL MANTRA FOR NETIZENS

Netizen = Internet user

Basic Cyber Hygiene Rules

1. Use strong passwords
2. Enable 2-factor authentication
3. Avoid suspicious links
4. Regular software updates
5. Install antivirus
6. Backup important data

Engineering View:
If system patches are not updated → vulnerability remains → attacker exploits.

7️⃣ CYBER OFFENSES

Now we study how crimes are executed technically.

7.1 HOW CRIMINALS PLAN ATTACKS

Attack Lifecycle (Step-by-step)

Step 1: Reconnaissance

Collect information about target.
Example:
Checking company website, LinkedIn employees.

Step 2: Scanning

Finding vulnerabilities.
Example:
Scanning open ports using tools.

Step 3: Gaining Access

Exploiting weakness.
Example:

Using SQL injection.

Step 4: Maintaining Access

Installing backdoor.

Step 5: Covering Tracks

Deleting logs.

7.2 SOCIAL ENGINEERING

Definition:
Psychological manipulation to trick users.

Types of Social Engineering

1. Phishing

Fake emails.

2. Spear Phishing

Targeted attack.

3. Vishing

Voice call scam.

4. Smishing

SMS scam.

5. Baiting

Infected USB drive.

Example:
Fake college fee payment portal capturing credentials.

7.3 CYBER STALKING

Definition:
Repeated online harassment or monitoring.
Methods:

- Spyware
- Fake accounts
- Threat emails

Legal Protection:
Covered under IT Act 2000.

7.4 CYBERCAFE AND CYBERCRIMES

Earlier criminals used cybercafes to hide identity.

Reasons:

- No strict ID proof
- Shared IP
- No tracking logs

Modern version:

- VPN
- TOR browser
- Proxy servers

7.5 BOTNETS – FUEL FOR CYBERCRIME

Definition

Botnet = Network of infected computers controlled remotely.

Each infected system = Bot or Zombie.

Working

1. Malware infects system.
2. Connects to command server.
3. Attacker sends commands.
4. All bots execute attack.

Types of Botnet Architecture

1. Centralized (Single control server)
2. Peer-to-peer (Decentralized)

Uses

- DDoS attack
- Spam emails
- Crypto mining
- Data theft

Example:

Thousands of infected computers attacking one website.

7.6 ATTACK VECTOR

Definition:
Method or path used to enter a system.

Types of Attack Vectors

1. Network Based

Open ports, weak firewall.

2. Application Based

SQL injection, XSS.

3. Human Based

Phishing.

4. Physical Based

USB malware.

Good. Difference tables are **very important for AKTU exams**.

Many 5–10 mark questions come directly from “Differentiate between…”.

Below are the **most important difference tables from Unit–1**, properly structured for exam writing.

1 Cybercrime vs Traditional Crime

Basis	Cybercrime	Traditional Crime
Medium	Computer / Internet	Physical world
Location	Borderless	Limited to specific location
Identity	Anonymous	Criminal usually visible
Speed	Very fast (seconds)	Slower
Evidence	Digital logs	Physical evidence
Example	Phishing, hacking	Theft, robbery

2 Computer as Target vs Computer as Tool

Basis	Computer as Target	Computer as Tool
Meaning	System itself attacked	System used to commit crime
Objective	Damage system	Perform illegal act
Example	Hacking server	Sending phishing emails
Damage Type	Data loss	Financial fraud
Technical Area	OS, Network Security	Social engineering

3 Information Security vs Cyber Security

Basis	Information Security	Cyber Security
Scope	Protects information (digital + physical)	Protects digital systems only
Focus	CIA triad	Network, systems, applications
Includes	Policies, procedures	Firewalls, IDS, encryption
Broader Concept	Yes	Subset of InfoSec
Example	Securing documents	Securing website

4 Confidentiality vs Integrity vs Availability (CIA)

Basis	Confidentiality	Integrity	Availability
Meaning	Prevent unauthorized access	Prevent unauthorized modification	Ensure access when required
Technique	Encryption	Hashing	Backup servers
Risk	Data leak	Data corruption	System crash
Example	Password protection	Digital signature	Cloud hosting

5 Hacker vs Cybercriminal

Basis	Hacker	Cybercriminal
Intention	May be ethical	Always illegal
Permission	Can have authorization	No authorization
Goal	Find weaknesses	Exploit for gain
Legal Status	Can be legal	Illegal
Example	Ethical hacker	Bank fraudster

6 Vulnerability vs Threat vs Risk vs Exploit

Term	Meaning	Example
Vulnerability	Weakness	Weak password
Threat	Possible danger	Hacker
Exploit	Tool/code used	Brute force script
Risk	Chance of damage	Account getting hacked

7 Phishing vs Spear Phishing vs Whaling

Basis	Phishing	Spear Phishing	Whaling
Target	Mass users	Specific person	High-level executives
Personalization	No	Yes	Highly personalized
Example	Fake bank email	Fake mail to professor	Fake mail to CEO

8 Botnet Centralized vs Peer-to-Peer

Basis	Centralized Botnet	P2P Botnet
Control	Single C&C server	Multiple nodes
Detection	Easier	Harder
Structure	Star topology	Distributed
Risk	Server shutdown stops botnet	Hard to stop
Example	IRC botnets	Advanced botnets

9 Attack Vector vs Vulnerability

Basis	Attack Vector	Vulnerability
Meaning	Path of attack	Weakness in system
Nature	External path	Internal flaw
Example	Email link	Weak password
Role	Used to enter	Exploited by attacker

10 Social Engineering vs Technical Attack

Basis	Social Engineering	Technical Attack
Target	Human psychology	System weakness
Skill Required	Communication	Technical knowledge
Example	Fake OTP call	SQL injection
Detection	Hard (human error)	Can be detected by IDS

1 1 Cyber Stalking vs Cyber Bullying

Basis	Cyber Stalking	Cyber Bullying

Nature	Repeated monitoring	Public harassment
Target	Specific person	Usually students
Intent	Obsession/control	Humiliation
Example	Tracking messages	Posting abusive comments

1️⃣2️⃣ DDoS vs DoS

Basis	DoS	DDoS
Full Form	Denial of Service	Distributed Denial of Service
Source	Single system	Multiple systems
Strength	Moderate	Very powerful
Example	One attacker flooding	Botnet attack

1️⃣3️⃣ Cybercrime vs Cyber Warfare

Basis	Cybercrime	Cyber Warfare
Motive	Financial	Political/National
Target	Individuals, companies	Government, military
Example	Ransomware	Attacking power grid

GKM